

Triangles

1. [June 2023]

Triangles ABC and DEF are congruent, where A corresponds to D , and B and E are right angles. The measure of angle A is 62° . What is the value of angle F ?

- A) 28° C) 90°
B) 62° D) 118°

2. [June 2023]

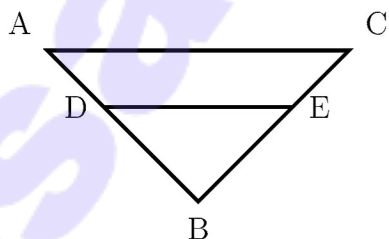
Triangle ABC is similar to triangle DEF , where angle A corresponds to angle D and angles C and F are right angles. The length $\overline{AB}$ is 2.4 times the length of $\overline{DE}$. If $\tan A = \frac{21}{20}$, what is the value of $\sin D$?

3. [June 2023]

In triangle RST , the measure of angle R is 10 degrees and the measure of angle T is 50 degrees. Point L lies on $\overline{RS}$, point K lies on $\overline{ST}$, and $\overline{LK}$ is parallel to $\overline{RT}$. What is the measure, in degrees, of angle SKL ? (Disregard the degree symbol when entering your answer.)

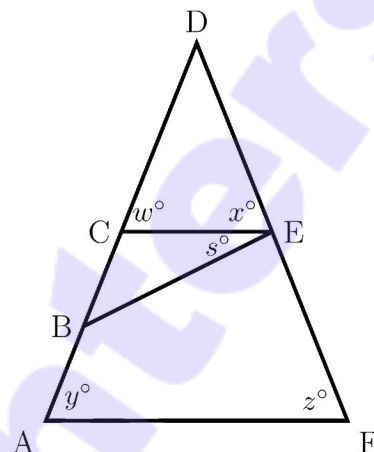
4. [August US 2023]

In the figure shown, line segments AC and DE are parallel. The measure of angle ACB is 58° , and $AB = 1.5DB$.



What is the measure, in degrees, of angle DEB ?
(Disregard the degree symbol when entering your answer.)

5. [August US 2023]



In figure shown, B and C lie on $\overline{AD}$, E lies on $\overline{DF}$, and $\overline{CE}$ is parallel to $\overline{AF}$. Which of the following is enough to prove that $\triangle BDE$ is similar to $\triangle EDC$?

- A) $w + y = x + z$
B) $w + x = y + z$
C) $s + x = z$
D) $s + x = y$

6. [August US 2023]

The area of a triangle is equal to x^2 square inches. The base of triangle is $5 + 2x$ inches, and the height of the triangle is $x - 2$ inches. What is the value of x ?

- A) 2.5 C) 5
B) 2.7 D) 10

7. [November 2023]

A triangle has a base length of 82 centimeters and a height of 164 centimeters. What is the area, in square centimeters, of the triangle?

- A) 246 C) 6,724
B) 3,362 D) 13,448

8. [December 2023]

In triangle RST , the measure of angle R is 11 degrees and the measure of angle T is 77 degrees. Point L lies on $\overline{RS}$, point K lies on $\overline{ST}$, and $\overline{LK}$ is parallel to $\overline{RT}$. What is the measure, in degrees, of angle SKL ? (Disregard the degree symbol when entering the answer.)

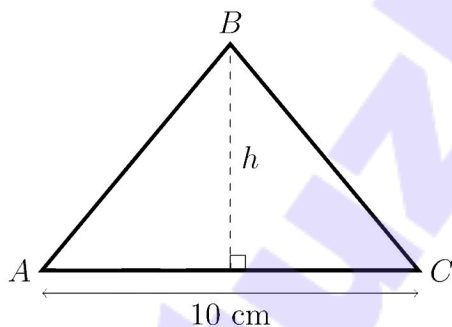
9. [December 2023]

The area of a triangle is equal to x^2 square centimeters. The length of the base of the triangle is $2x + 16$ centimeters, and the height of the triangle $x - 6$ centimeters. What is the value of x ?

10. [March 2024]

Triangles ABC and DEF are congruent, where A corresponds to D , and B and E are right angles. The measure of angle A is 27° . What is the measure, in degrees, of angle F ?

11. [March 2024]



The area of triangle ABC is 30 square centimeters. What is the height h , in centimeters, of this triangle?

- | | |
|------|-------|
| A) 3 | C) 10 |
| B) 6 | D) 12 |

12. [May US 2024]

Triangles ABC and DEF are congruent, where A corresponds to D , and D , and B and E are right angles. The measure of angle A is 71° . What is the measure, in degrees, of angle F ?

13. [March 2024]

In right triangle RST , the sum of the measures of angle R and angle S is 90 degrees. The value of $\sin(R)$ is $\frac{4\sqrt{2}}{9}$. What is the value of $\cos(S)$?

- | | |
|--------------------------|--------------------------|
| A) $\frac{4\sqrt{2}}{9}$ | C) $\frac{7\sqrt{2}}{8}$ |
| B) $\frac{4\sqrt{2}}{7}$ | D) $\frac{9\sqrt{2}}{8}$ |

14. [March 2024]

In $\triangle ABC$, $\angle B$ is a right angle and the length of $\overline{BC}$ is 248 millimeters. If $\cos A = \frac{3}{5}$, what is the length, in millimeters, of $\overline{AB}$?

- | | |
|--------|--------|
| A) 62 | C) 248 |
| B) 186 | D) 310 |

15. [May 2024]

In triangle ABC and triangle DEF , sides AB and DE each have a side length of 6 inches, and angles A and D each have an angle measure of 40° . Which of the following additional pieces of information is(are) sufficient to prove whether triangle ABC is congruent to triangle DEF ?

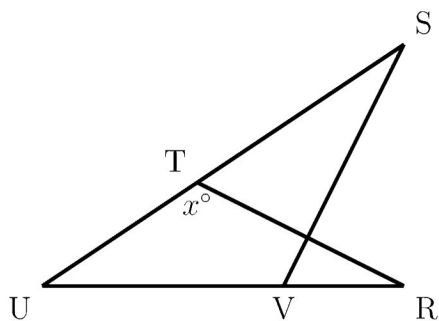
- I. The measures of angles B and C are equal.
- II. The lengths of sides AC and DF are equal.
- III. The lengths of sides BC and EF are equal.

- A) I is sufficient, but II and III are not.
- B) II is sufficient, but I and III are not.
- C) III is sufficient, but I and II are not.
- D) I is sufficient and III is sufficient, but I is not.

16. [May US 2024]

Triangles ABC and DEF are similar. Each side length of triangle ABC is 2 times the corresponding side length of triangle DEF . The area of triangle ABC is 300 square centimeters. What is the area, in square centimeters, of triangle DEF ?

17. [May 2024]



Note: Figure not drawn to scale

In the figure, $RT = TU$, the measure of angle VST is 23° , and the measure of angle RVS is 34° . What is the value of x ?

18. [May 2024]

In isosceles triangle ABC , sides AB and AC are congruent. Point D divides BC such that the length of $\overline{BD}$ is $\frac{5}{7}$ of the length of $\overline{BC}$. Point E lies on side AB and point F lies on side AC such that when segments DE and DF are drawn, angle BED is congruent to angle CFD . If the length of $\overline{BE}$ is 10, what is the length of $\overline{CF}$?

- | | |
|-------|-------|
| A) 4 | C) 50 |
| B) 14 | D) 70 |

19. [May US 2024]

In a triangle ABC , the measure of angle A is 48° , the measure of angle B is 90° , and the measure of angle C is $(\frac{k}{2})^\circ$. What is the value of k ?

- | | |
|-------|-------|
| A) 42 | C) 84 |
| B) 45 | D) 96 |

20. [May US 2024]

A perimeter of an isosceles right triangle is $38 + 38\sqrt{2}$ inches. What is the length, in inches, of the hypotenuse of this triangle?

- | | |
|-----------------|-----------------|
| A) 19 | C) 38 |
| B) $19\sqrt{2}$ | D) $38\sqrt{2}$ |

21. [June 2024]

In right triangle ABC , angles A and B are acute, side AC has a length of 21.7, and $\tan B = \frac{1}{8}$. What is the length of side BC , rounded to the nearest tenth?

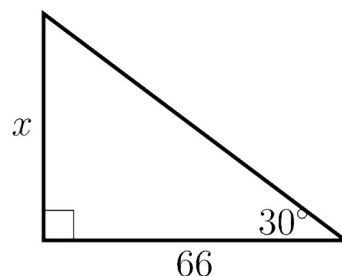
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|----------|--------|
| A) 470.9 | C) 4.7 |
| B) 173.6 | D) 2.7 |

22. [June 2024]

In right triangle ABC , angles A and B are acute. Side AC has a length of 21.2, and $\tan B = \frac{1}{6}$. What is the length of side BC , rounded to the nearest tenth?

- | | |
|--------|----------|
| A) 3.5 | C) 127.2 |
| B) 4.6 | D) 449.4 |

23. [June 2024]



Note: Figure not drawn to scale. In the triangle shown, what is the value of x ?

- | | |
|-----------------|-----------------|
| A) 22 | C) $66\sqrt{3}$ |
| B) $22\sqrt{3}$ | D) 132 |

24. [June 2024]

For triangle PQR and triangle STU , angles P and S each measure 30° . $QR = 16$ and $TU = 12$. Which additional piece(s) of information is(are) sufficient to prove that triangle PQR is similar to triangle STU ?

- I. The length of PQ is $\frac{1}{3}$ the length of ST .
 - II. The length of PR is $\frac{1}{3}$ the length of SU .
- A) I is sufficient, but II is not.
 B) II is sufficient, but I is not.
 C) I is sufficient, and II is sufficient.
 D) Neither I nor II is sufficient.

25. [August 2024]

Triangle QRS is similar to triangle FGH such that Q , R , and S correspond to F , G , and H , respectively. Each side of triangle FGH has $\frac{1}{4}$ the length of its corresponding side in triangle QRS . In triangle FGH , the measure of angle F is 35° , the measure of angle G is 8° , and $FG = a$. In triangle QRS , the measure of angle Q is b° and $QR = 20$. What is the value of $a + b$?

26. [August 2024]

The area of a triangle is equal to x^2 square centimeters. The length of the base of the triangle is $2x + 12.5$ centimeters, and the height of the triangle is $x - 5$ centimeters. What is the value of x ?

27. [August 2024]

In triangle FGH and triangle KLM , the measures of angles G and L are each 60° . The lengths of GH and LM are each 34 centimeters, and $\frac{GH}{GF} = \frac{LM}{LK}$. Which additional piece of information would be necessary to prove that triangle FGH is congruent to triangle KLM ?

- A) The lengths of FG and KM are each 17 centimeters.
- B) The measures of angles H and M are each 30° .
- C) The measures of angles F and K are each 90° .
- D) No additional information is necessary.

28. [August US 2024]

Triangles ABC and DEF are similar, where A corresponds to D , and B corresponds to E . The measure of angle A is 26° , and $AB = 4$. Which of the following statements must be true?

I. The measure of angle D is 26°

II. $DE = 4$

- | | |
|------------|---------------------|
| A) I only | A) I and II |
| A) II only | A) Neither I nor II |

29. [August 2024]

If the perimeter of an equilateral triangle with a height of 46 units is $a\sqrt{3}$ units, what is the value of a ?

- | | |
|-------|----------------------|
| A) 46 | C) $\frac{2,116}{3}$ |
| B) 92 | D) $\frac{4,232}{3}$ |

30. [August US 2024]

A triangle inequality theorem states that the sum of any two sides of a triangle must be greater than the length of the third side. If a triangle has side length of 8 and 13, which of inequality represents the possible length, x , of the third side of the triangle?

- A) $x < 21$
- B) $x > 21$
- C) $5 < x < 21$
- D) $x < 5$ or $x > 21$

31. [August 2024]

The perimeter of an equilateral triangle is 642 centimeters. The three vertices of the triangle lie on a circle. The radius of the circle is $w\sqrt{3}$ centimeters. What is the value of w ?

32. [October 2024]

The perimeter of an isosceles right triangle is $18 + 18\sqrt{2}$ inches. What is the length, in inches, of the hypotenuse of this triangle?

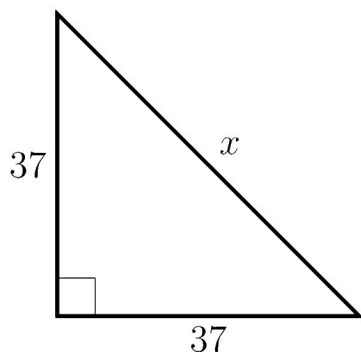
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|----------------|-----------------|
| A) 9 | C) 18 |
| B) $9\sqrt{2}$ | D) $18\sqrt{2}$ |

33. [October US 2024]

In $\triangle FGH$, the measure of $\angle F$ and $\angle G$ are each 39° . What is the measure of $\angle H$?

- | | |
|----------------|----------------|
| A) 78° | C) 129° |
| B) 102° | D) 141° |

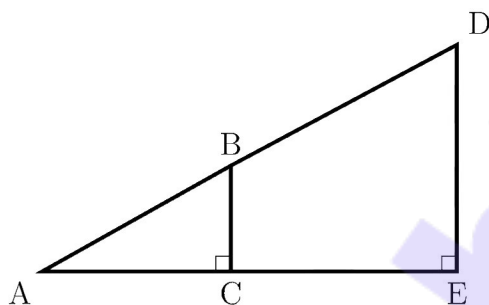
34. [October US 2024]



In the right triangle shown, what is the value of x ?

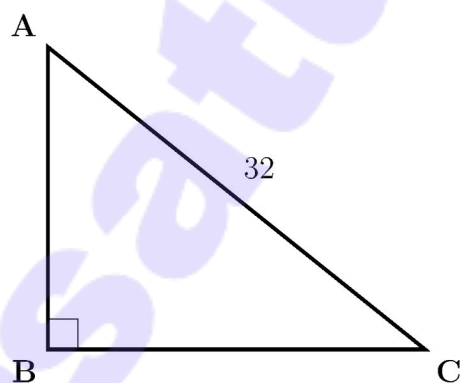
- A) $\sqrt{2}$ A) $\sqrt{74}$
 A) $\sqrt{37}$ A) $37\sqrt{2}$

35. [October US 2024]



In the figure shown $AB = \sqrt{89}$ units, $AC = 8$ units, and $CE = 24$ units. What is the area, in square units, of triangle ADE ?

36. [November 2024]



For triangle ABC , the length of AB is 11 less than the length of AC . Point D (not shown) lies on AC such that BD (not shown) is perpendicular to AC . What is the value of $\frac{BC}{BD}$?

37. [November 2024]

In triangle ABC , angle B is a right angle. The length of side AB is $10\sqrt{39}$ and the length of side BC is $24\sqrt{39}$. What is the length of side AC ?

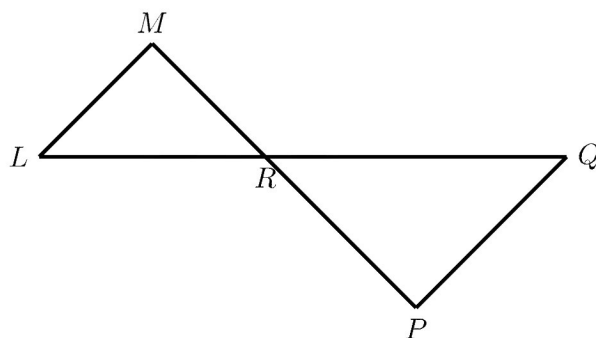
- A) $14\sqrt{39}$ C) $34\sqrt{39}$
 B) $26\sqrt{39}$ D) $\sqrt{24 \cdot 39}$

38. [November US 2024]

In triangle ABC and triangle DEF , AB and DE are each equal to 5 centimeters, and angles A and D each have measure 35° . Which additional piece of information is sufficient to prove that triangle ABC is congruent to triangle DEF ?

- A) The lengths of sides BC and EF are equal.
 B) The lengths of sides AC and DF are equal.
 C) The measures of angles B and C are equal.
 D) No additional information is necessary to prove that the two triangles are congruent.

39. [November US 2024]



Note: Figure not drawn to scale

In the figure, LQ intersects MP at point R , and $\overline{LM}$ is parallel to $\overline{PQ}$. The lengths of MR , LR , and RP are 9, 10, and 19, respectively. What is the length of LQ ?

- A) $\frac{280}{19}$ C) $\frac{271}{9}$
 B) $\frac{190}{9}$ D) $\frac{280}{9}$

40. [November US 2024]

The table gives the perimeters of similar triangles TUV and XYZ , where TU corresponds to XY . The length of TU is 16.

	Perimeter
Triangle TUV	44
Triangle XYZ	352

What is the length of XY ?

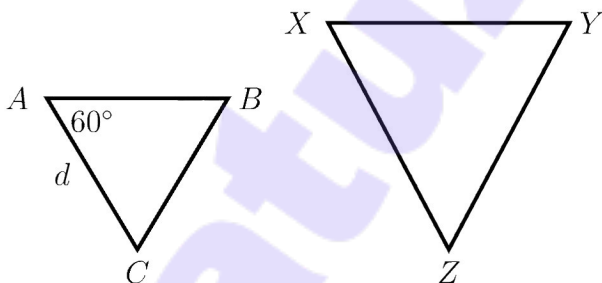
- A) 2
- C) 60
- B) 16
- D) 128

41. [November US 2024]

The area of a triangle is 270 square centimeters. The length of the base of the triangle is 12 centimeters greater than the height of the triangle. What is the height, in centimeters, of the triangle?

- A) 15
- C) 30
- B) 18
- D) 36

42. [December US 2024]

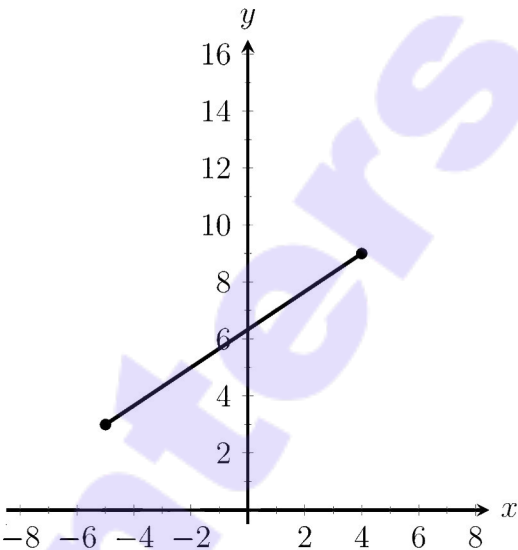


Note: Figure not drawn to scale

For the triangles shown, triangle ABC dilated by a factor of 5 to obtain triangle XYZ , where $d = 19$. What is the measure, in degrees, of angle X ?

- A) 12
- C) 60
- B) 55
- D) 65

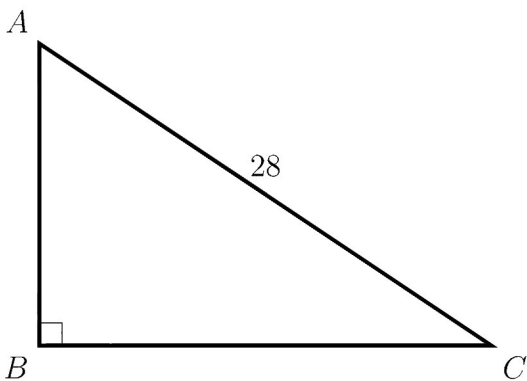
43. [November US 2024]



The line segment shown in the xy -plane represents one of the legs of a right triangle. The area of this triangle is $48\sqrt{13}$ square units. What is the length, in units, of the other leg of this triangle?

- A) 16
- C) $3\sqrt{13}$
- B) 32
- D) 24

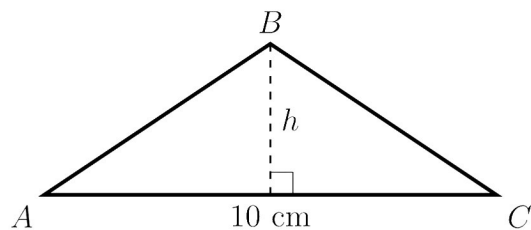
44. [December US 2024]



Note: Figure not drawn to scale

For triangle ABC , the length of $\overline{AB}$ is 11 less than the length of $\overline{AC}$. Point D (not shown) lies on $\overline{AC}$ such that BD (not shown) is perpendicular to $\overline{AC}$. What is the value of $\frac{BC}{BD}$?

45. [December US 2024]

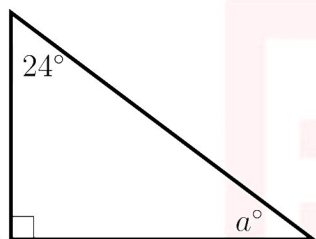


Note: Figure not drawn to scale

The area of triangle ABC is 220 square centimeters. What is the height h , in centimeters, of this triangle?

- A) 10
- B) 22
- C) 44
- D) 88

46. [December 2024]

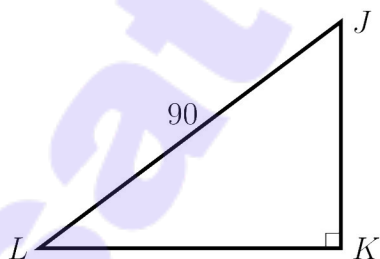


Note: Figure not drawn to scale

In the right triangle shown, what is the value of a ?

- A) 24
- B) 66
- C) 90

47. [December 2024]

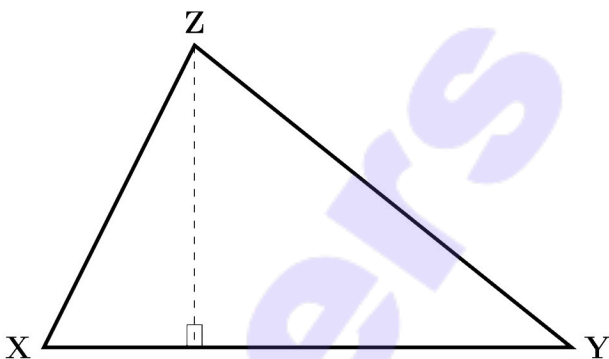


Note: Figure not drawn to scale

In right triangle JKL , the tangent of $\angle L$ is $\frac{3}{4}$. What is the length of JK ?

- A) $\frac{35}{16}$
- B) $\frac{15}{4}$
- C) 54
- C) 72

48. [December 2024]

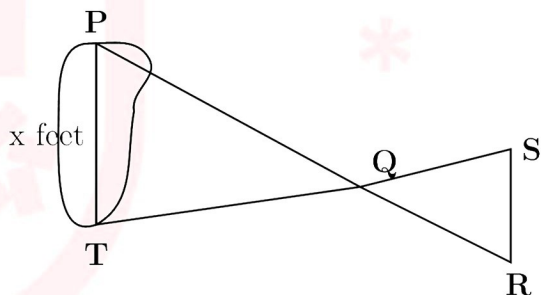


Note: Figure not drawn to scale.

In the figure shown, the measure of angle X is 52° . The length of $\overline{XY}$ is 24 units and the length of $\overline{XZ}$ is 17 units. What is the, in square units, of triangle XYZ ?

- A) 204
- B) 408
- C) $204\sin 52^\circ$
- D) $408\sin 52^\circ$

49. [December 2024]



Note: Figure not drawn to scale.

A property owner wants to find the length x , in feet, across a pond as represented in the figure. PR intersects ST at point Q , and $\angle PTQ$ is congruent to $\angle RSQ$. The length represented by PQ , TQ , QS , and RS were determined to be 4,200 feet, 8,400 feet, 1,400 feet, and 1,600 feet, respectively. What is the value of x ?